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30 JULY 1954
SUPERSEDING
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MILITARY SPECIFICATION

FUZES, BOMB, MECHANICAL TIME, AN-M145E3, AN-M146E3, AN-M147E3, M152E3, AND M155E3; PARTS FOR

This specification has been approved by the Department of Defense for use of the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 *This specification covers parts for the following fuzes:*

AN-M145E3 — Fuze, Bomb, Nose, Mechanical Time
AN-M146E3 — Fuze, Bomb, Nose, Mechanical Time
AN-M147E3 — Fuze, Bomb, Nose, Mechanical Time
M152E3 — Fuze, Bomb, Tail, Mechanical Time
M155E3 — Fuze, Bomb, Nose, Mechanical Time

2. APPLICABLE DOCUMENTS

2.1 The following specifications, drawings and publications, and all drawings, standards and specifications referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-P-116 — Preservation, Methods of
MIL-G-2550 — General Specification for Ammunition Except Small Arms Ammunition.
MIL-P-10025 — Packing and Marking For Domestic Shipment of Inert Ammunition Components; General Specification For.

DRAWINGS

ORDNANCE CORPS

P-83708 — Fuze, Bomb, Tail, Mechanical Time, M152E3 Assembly and Details.
P-83745 — Fuze, Bomb, Nose, Mechanical Time, AN-M146E3 Assembly and Details.
P-83815 — Fuze, Bomb, Nose, Mechanical Time, M155E3 Assembly and Details.
P-84190 — Fuze, Bomb, Nose, Mechanical Time, AN-M147E3 Assembly and Details.
P-84247 — Fuze, Bomb, Nose, Mechanical Time, AN-M145E3 Assembly and Details.
81-3-127 — Fixture, Testing, Fuze Timing

PUBLICATIONS

ORDNANCE CORPS

ORD-SIP-M7-255 — Fuzes, Bomb, Mechanical Time: M155E3 Parts

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ORD–SIP–M7–256 — Fuzes, Bomb, Mechanical Time: AN–M145E3, AN–M146E3, AN–M147E3, and M152E3 Parts.

(Copies of drawings, specifications, standards and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—Material and parts shall be in accordance with applicable drawings and specifications.

3.2 The parts shall comply with all the requirements specified on drawings and in applicable specifications.

3.3 Torque (not applicable to Fuze M155E3). — The metal parts assembly shall withstand the torque test specified in 4.4

3.4 Functioning and freeness of action. — The metal parts assembly shall withstand the functioning and freeness of action test specified in 4.5.

3.5 Spin back.—The metal parts assembly shall withstand the spring back test specified in 4.6.

3.6 Static functioning.—Fuzes shall function within ±1 second of the set time of the fuze. The sliders shall arm in not less than 3 seconds nor more than 4.75 seconds after ejection of the safety pin (see 6.2).

3.7 Workmanship.—All parts shall be free of chips, dirt, grease, rust and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.—A lot shall consist of all fuze metal parts manufactured by one manufacturer under one contract, in one unchanged process, in accordance with same drawing, same drawing revision, same specification and same specification revision.

4.2 Sampling. — Number of samples, acceptance and rejection criteria used for the determination of lot acceptance for tests as specified herein, shall be in accordance with Standard Inspection Procedure ORD–SIP–M7–255 and ORD–SIP–M7–256.

4.3 Inspection. — Inspection shall be as specified in Specification MIL–G–2550.

4.4 Torque test for metal parts assemblies (not applicable to Fuze M155E3).

4.4.1 A torque test shall be conducted on sample assemblies to determine the satisfactory engagement of pinion and gears. Apply a torque which will tend to rotate the arming sleeve with respect to the arming hub. The arming sleeve shall be gripped in a fixture by smooth steel jaws which are shaped to fit the outside diameter of the arming sleeve. The jaws shall apply a gripping pressure of 3 to 4 pounds on the arming sleeve, which will result in a frictional torque, to prevent its rotation, of approximately 0.4 inch-pounds. The assembly shall hang freely from the jaws of the testing device and, while the body is restrained from turning, the vane assembly shall be rotated for approximately one full turn. Observe for satisfactory engagement of pinion and gears.

4.4.2 In a satisfactory assembly, rotation of the vane will result in the steady turning of the arming sleeve, in the jaws of the fixture, against the uniform friction torque of the jaws which will tend to restrain the rotation of the vane.

4.4.2. In an assembly with a gear or pinion missing, the arming sleeve will not turn in the jaws of the fixture, but will turn with respect to the arming hub, and there will be no restraint on the turning of the vane. In an assembly in which the gears and pinion

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do not engage properly, on turning of the vane, the sleeve will turn intermittently, or not at all, with respect to the jaws and consequently, will turn with respect to the arming hub. This will be accompanied with a jerky action as the teeth of the pinion jump the teeth of the gears.

4.5 Test for functioning and freeness of action of the arming mechanism.

4.5.1 *AN–M145E3, AN–M146E3, AN–M147E3 and M152E3 Fuzes.* — The arming mechanism of the metal parts assemblies shall function properly and freely. Proper functioning is indicated by the down and up movement of the arming sleeve. The freeness required is such that when an acceptable metal parts assembly is placed in a wind stream of 45 miles per hour, the vane should rotate at a speed of not less than 1200 rpm. The metal parts assembly shall be tested in an approved device in which the arming vane is rotated in the direction to arm from 60 to 90 complete revolutions, and then returned to the original position. The down and up movement of the arming sleeve with respect to the arming hub shall be gaged or observed during this test to determine that the arming mechanism is functioning. The operation of the arming mechanism shall also be observed to determine that it is free from undue friction.

4.5.2 *M155E3 Fuze.*—This test shall be conducted in the same manner as in 4.5.1 with the following exceptions:

(1) The rotation in the direction to arm shall be 4 to 6 turns instead of 60 to 90.

(2) The movement down and up shall be .125 inches per 4 turns instead of .05 inches per 60 turns.

4.6 Spin back test for metal parts assemblies. — Sample metal parts assemblies shall be given a spin back test. Looking at the vane end of the metal parts assembly, spin the vane counterclockwise by tapping it lightly with the finger until the vane comes to an abrupt stop indicating that the movable gear has come in contact with the stationary gear. Spin the vane in a clockwise direction four or five turns, and then counterclockwise until the gears are again in contact and the vane rotation stops. In no instance is the vane to be forced back past the point where the vane rotation stops. The vane is then to be backed off or pre-armed 17 turns, (except for M155E3 fuzes which are to be pre-armed only one-half to one turn with the fuze set at the 5 second setting). All fuze types are to be turned through their entire setting range to be certain that the fuzes are correctly pre-armed. A defective assembly is one in which the vane can be continuously spun counterclockwise.

4.7 Static functioning test.—Sample fuzes shall be selected, divided into three groups, and conditioned at the following temperatures –65° F. ±5° F., 75° F. ±10° F. and 125° F. ±5° F., until the internal temperature of the fuzes is within the specified temperature range. One third of each of the groups of fuzes, conditioned as above, shall be tested at each of the following time settings: 6, 45 and 90 seconds. Sample M155E3 fuzes shall be tested in a fixture as shown on figure 1; sample AN–M145E3, AN–M146E3, M147E3 and M152E3 fuzes shall be tested in a fixture as shown on Drawing 81–3–127.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.—Packaging and packing shall be in accordance with Method 1A–2 of Specification MIL–P–116.

5.2 Marking.—Marking for shipment shall be in accordance with Specification MIL–P–10025.

5.3 Data cards.—Data card information shall be in accordance with Specification MIL–G–2550.

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6. NOTES

6.1 Precautions should be taken against introduction of moisture into the striker and arming mechanism when fuzes are subjected to –65°F., and subsequently removed and dried.

6.2 Failure of the slider to function due to malfunctioning of the movement (i.e., failure of the movement to start to run for 6 seconds) should not be considered as a slider malfunction.

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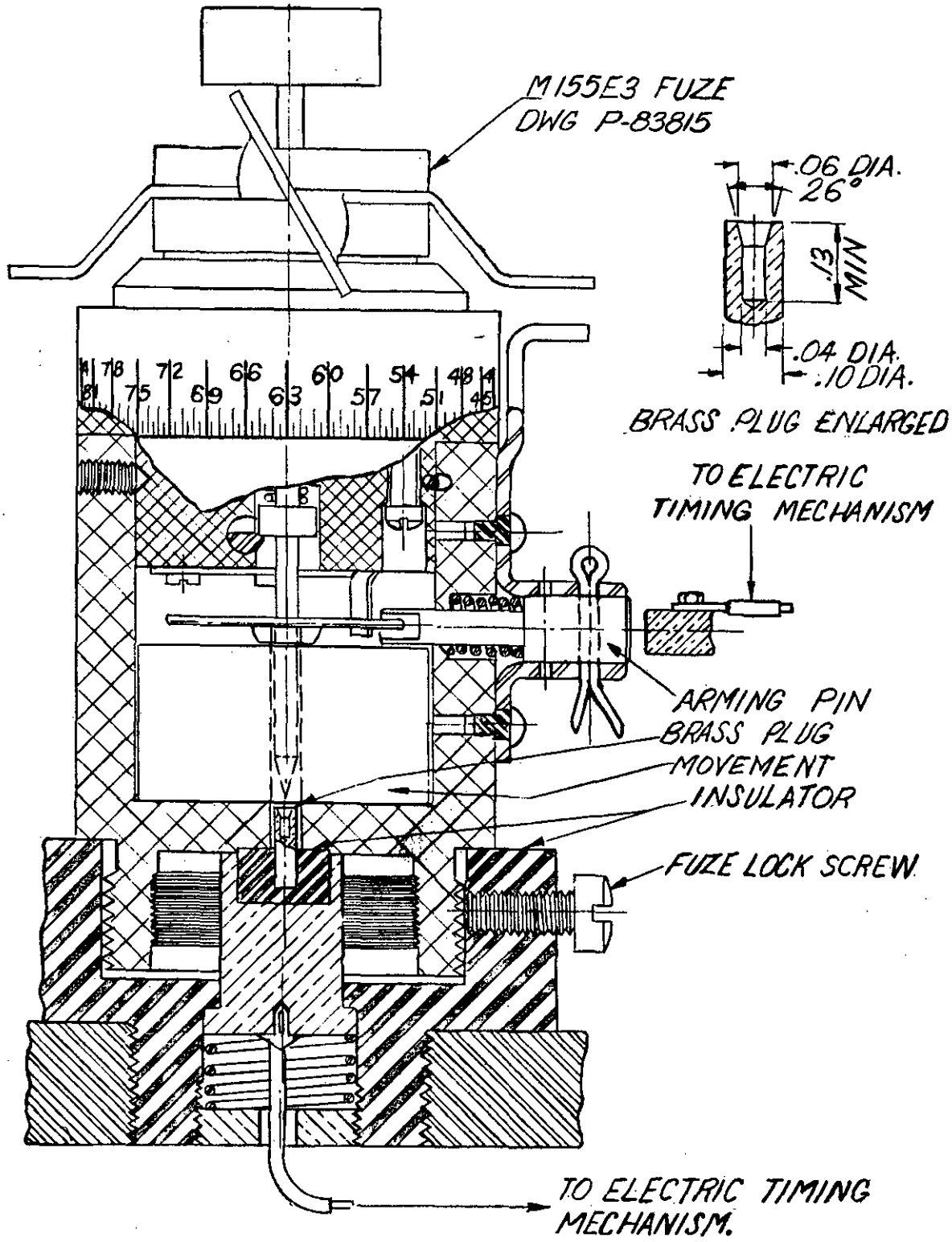


FIGURE 1.

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